

30 *Use of the Data Booklet is relevant to this question.*

When 0.95 g of a chromium compound is first dissolved in water and treated with an excess of aqueous silver nitrate solution, 0.50 g of white precipitate was collected.

Given that the general formula for the chromium compound is $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$, what is the formula of the chromium-containing ion?

- A** Cr^{3+} **B** $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ **C** $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]^{2+}$ **D** $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]^+$